KAIZEN MATHS

Lesson starter

Discrete Random Variables and Probability Distributions

Standard Distributions

Name

S1

X ~ B(8, 0.60). Find E(X), Var(X), and P(X=1).

S2

X is geometric with p = 0.50. Find E(X), Var(X), and P(X=3).

S3

X ~ Poisson(5). Find E(X), Var(X), and P(X=1).

S4

Identify the distribution:
"Count emails arriving between 9:00 and 10:00"

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Show your working

KAIZEN MATHS

Exit ticket

Discrete Random Variables and Probability Distributions

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Name

E1

X ~ B(6, 0.60). Find E(X), Var(X), and P(X=3).

E2

X is geometric with p = 0.20. Find E(X), Var(X), and P(X=5).

E3

X ~ Poisson(3). Find E(X), Var(X), and P(X=0).

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Teacher answers & working

Discrete Random Variables and Probability Distributions · Standard Distributions

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S1

X ~ B(8, 0.60). Find E(X), Var(X), and P(X=1).

Answer:

E(X)=4.8, Var(X)=1.92, P(X=1)=0.008

Working:

Use binomial formulas

$E(X) = np = 8(0.60) = 4.8$

$Var(X) = np(1-p) = 1.92$

Use $P(X=k) = nCk\ p^k(1-p)^{(n-k)}$

$P(X=1) = 8C1(0.60)^1(0.40)^7 = 0.008$

0.008

S2

X is geometric with p = 0.50. Find E(X), Var(X), and P(X=3).

Answer:

E(X)=2.0, Var(X)=2.00, P(X=3)=0.125

Working:

Use geometric formulas

$E(X)=1/p=2.0$

$Var(X)=(1-p)/p^2=2.00$

First success on trial k

$P(X=3) = (1-p)^2p = 0.125$

0.125

S3

X ~ Poisson(5). Find E(X), Var(X), and P(X=1).

Answer:

E(X)=5, Var(X)=5, P(X=1)=0.034

Working:

Use Poisson properties

E(X)=lambda=5 and Var(X)=lambda=5

Use P(X=k)=e^-lambda $lambda^k/k!$

$P(X=1) = e^{-5} \times 5^1 / 1! = 0.034$

0.034

Teacher answers & working

Discrete Random Variables and Probability Distributions · Standard Distributions

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S4

Identify the distribution:
"Count emails arriving between 9:00 and 10:00"

Answer:

Poisson (lambda = mean emails per hour)

Working:

Identify the structure

counting events in a fixed time interval

Name the distribution and parameter

Poisson distribution with lambda = mean emails per hour

Poisson

E1

X ~ B(6, 0.60). Find E(X), Var(X), and P(X=3).

Answer:

E(X)=3.6, Var(X)=1.44, P(X=3)=0.276

Working:

Use binomial formulas

$E(X) = np = 6(0.60) = 3.6$

$Var(X) = np(1-p) = 1.44$

Use $P(X=k) = nCk\ p^k(1-p)^{(n-k)}$

$P(X=3) = 6C3(0.60)^3(0.40)^3 = 0.276$

0.276

E2

X is geometric with p = 0.20. Find E(X), Var(X), and P(X=5).

Answer:

E(X)=5.0, Var(X)=20.00, P(X=5)=0.082

Working:

Use geometric formulas

$E(X)=1/p=5.0$

$Var(X)=(1-p)/p^2=20.00$

First success on trial k

$P(X=5) = (1-p)^4p = 0.082$

0.082

Teacher answers & working

Discrete Random Variables and Probability Distributions · Standard Distributions

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E3

X ~ Poisson(3). Find E(X), Var(X), and P(X=0).

Answer:

E(X)=3, Var(X)=3, P(X=0)=0.050

Working:

Use Poisson properties

E(X)=lambda=3 and Var(X)=lambda=3

Use P(X=k)=e^-lambda lambda^k/k!

$P(X=0) = e^{-3} \times 3^0 / 0! = 0.050$

0.050